

VisDev: A Stroke Deviation Visualization System Using Point-Set Registration for Handwriting Training

Yuzheng Xu
OMRON SINICX Corporation
Tokyo, Japan
yuzheng.xu@sinicx.com

Atsushi Hashimoto
OMRON SINICX Corporation
Tokyo, Japan
atsushi.hashimoto@sinicx.com

Shigeo Yoshida
OMRON SINICX Corporation
Tokyo, Japan
shigeo.yoshida@sinicx.com

Yoshitaka Ushiku
OMRON SINICX Corporation
Tokyo, Japan
yoshitaka.ushiku@sinicx.com

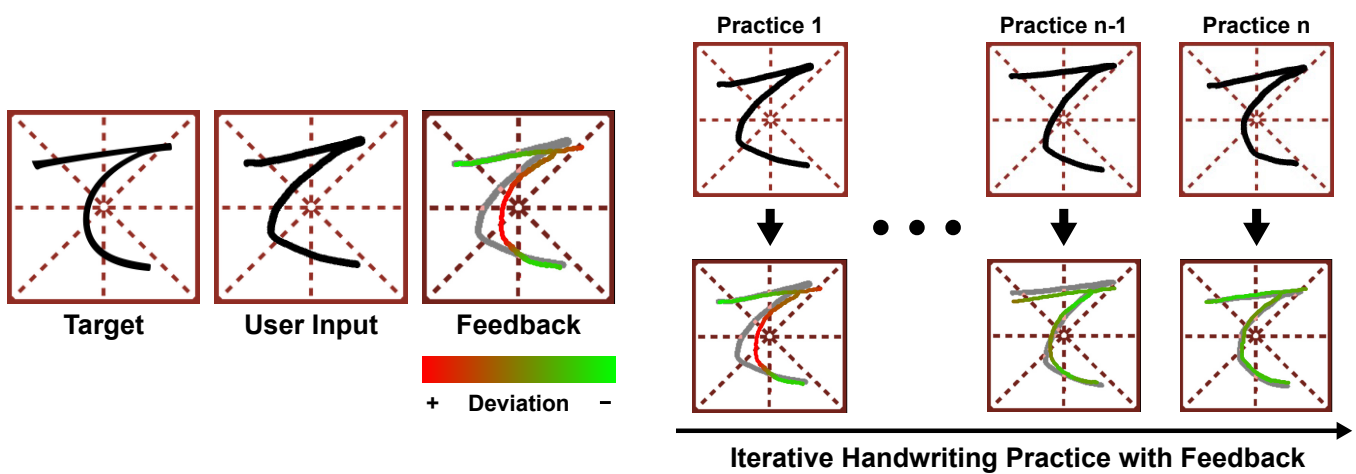


Figure 1: The user writes based on the target characters and receives animated feedback. Different colors in the feedback represent the deviation between the input and the target. By repeating this process, the user can gradually improve their handwriting.

Abstract

Current handwriting training methods, such as tracing and copying, either lack detailed, immediate feedback or place minimal demands on memory, limiting users' ability to iteratively correct mistakes and improve handwriting quality. To address this, we introduce VisDev, a system that uses a point-set registration method to align and visualize deviations between sequences of points (strokes) in handwritten and target characters, providing targeted, interactive feedback. By highlighting correct and incorrect strokes based on its point-wise correspondence, VisDev enables users to identify and correct specific inaccuracies without losing what has already been done correctly. It requires users to recall previous deviations, creating an iterative learning process that improves accuracy over

time. We evaluated VisDev in a user study, showing improvement in handwriting accuracy after 5 training sessions compared to traditional methods. Our work bridges the gap between traditional handwriting education and digital platforms, offering a solution for improving handwriting through immediate feedback.

CCS Concepts

• **Human-centered computing** → **Information visualization**; **Interactive systems and tools**.

Keywords

Hand Writing Training, Adaptive Feedback Systems

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1 Introduction

Tracing and copying are fundamental in handwriting education, especially in countries with traditions of Chinese character-based education [4, 31]. Additionally, calligraphy is an essential part of writing education in these cultures, where both tracing and copying methods are often used to pursue the aesthetics of writing. This educational background has been shown to influence people's visuo-motor ability [29]. Therefore, it is foreseeable that adults with a background in character-based education are more inclined to use tracing and copying methods to learn calligraphy or new characters. We aim to design a tool for adults with this educational background that can quickly enhance their writing ability. Since they are already familiar with tracing and copying, it would be ideal to combine the strengths of both methods.

Tracing provides precise feedback but lacks the complexity and challenges present in real writing scenes. Copying demands memory but does not highlight errors. The effectiveness of these two methods has long been a research topic [2, 15, 27], though conclusions have been varied and, at times, contradictory [6]. Moreover, previous studies often lacked kinematic analysis due to the limitations of the technology available at the time [6]. Effectively combining the advantages of tracing and copying, and providing an objective evaluation, is a valuable direction for research.

We propose *VisDev* to integrate tracing and copying by visualizing handwriting deviations. Users can strive to memorize characters like copying, also be provided detailed feedback similar to tracing. We guide users through a set process where they start with the copying, then visualize errors and iteratively repeat attempts via a tracing way over a short period. The process is shown in Figure 1. It also illustrates the changes in writing during the use of the system.

For the contributions, first, we merge copying with tracing-like feedback, enhancing learning by combining memorization and precise guidance. Second, we visualize handwriting deviations using point-set registration and highlight them through color and animation for clearer recognition. Third, we conduct a preliminary experiment using our method compared to the tracing baseline. Both methods are effective, but ours is superior in both structure and relative positions.

2 Related Work

2.1 Mechanisms and Approaches for Writing Training

Writing training research has a long history [2, 11, 15, 23, 24, 26, 27], demonstrating that instruction improves learning [17, 27]. Søvik [24] outlines a three-step feedback process: attempt, compare, and adjust, highlighting feedback's importance. Copying requires memorization and offers limited feedback, making quality assessment difficult. In contrast, tracing provides immediate feedback by reducing cognitive load. While studies comparing copying and tracing have mixed findings [2, 6, 15, 27], recent work suggests tracing may benefit adult learners in the short term [6]. Because our target users are adults, we adopt tracing as our baseline without directly comparing the two methods.

2.2 Feedback for Writing Training

Visual, auditory, and tactile feedback improve handwriting training. Persketchtivity [18] focuses on visual feedback for sketching, while RealPen [5] and SpARKlingPaper [10] replicate paper-like writing with sound or augmented surfaces. PenSight [21] adds a camera to analyze writing, and LightWrite [28] uses sound to guide blind or low-vision (BLV) users. Haptic feedback like electromagnetic guidance [16] also helps improvement. Among these, visual feedback is the most cost-effective and exists in traditional writing. By using visual feedback, our approach allows seamless skill transfer from paper to tablet.

2.3 Sketch Training and Visualization

Many works optimize sketch instruction through visualization methods. Persketchtivity [18] highlights correct and incorrect areas, while FluidSketches [1] morphs strokes into ideal forms. iCanDraw [9] iteratively refines face sketches, and Drawing Assistant [12] extracts construction lines and provides real-time feedback. EasySketch2 [14] helps children improve fine motor skills with personalized feedback. Berney and Bétrancourt [3] show that animated feedback enhances learning. Adopting a similar iterative approach, we focus on proficient writers, not beginners like Kanji Workbook [25], and aim to improve handwriting's aesthetic quality through tracing and copying of ideal characters.

3 Method

3.1 Design Considerations

Our primary goal is to deliver feedback that clearly highlights the deviations between the user's writing and the target. Overly complex visual presentations can overwhelm users, so we focus on providing informative, easy-to-understand feedback. The target users are adults familiar with East Asian scripts who want to refine their handwriting aesthetics. Using a copying-based approach, we first allow users to observe the character and then attempt to copy it. Subsequently, through error visualization, users can understand the differences between their writing and the ideal template, mimicking the tracing process. This way, users can experience both copying and tracing. To show the deviations between written and target samples, given the vast variety of East Asian characters and styles, we employ a simple, versatile solution that does not require additional training for each character.

For the feedback mechanism, we adopt retrospective feedback, where users receive feedback immediately after completing their writing. In contrast, real-time feedback provides corrections during writing. However, to maintain a natural handwriting flow without interruptions, we prioritize retrospective feedback as the better approach.

3.2 Correspondence from Point-set Registration

Point-set registration [20] aligns two point sets by determining transformation parameters that minimize their differences. We thus use it to find the correspondence between written and target characters. It does not require extensive pre-processing of the template itself, meeting the diverse needs of our target users. Transformations fall into rigid (rotation, translation) and non-rigid (adaptable

deformations) categories. Our research uses both: non-rigid registration aligns characters with targets, and rigid registration assesses results. We employ Coherent Point Drift (CPD) [22], known for its robustness against noise, outliers, and varying sizes of point sets. We implement CPD using the probreg library [13]. To apply it, character images are skeletonized via the Zhang-Suen algorithm [30], then converted into 2D point clouds. We set a maximum of 50 iterations and tolerances of 0.000001 (non-rigid) and 0.0001 (rigid). The correspondences are used for the following feedback animation.

3.3 Feedback Animation

To provide clear and informative feedback from the registration results, we adopted two complementary strategies. First, we implemented a color gradient on the target character, ranging from green (correct) to red (incorrect), to immediately convey accuracy at a glance. It enables users to quickly identify areas of high and low accuracy without additional cognitive effort. Second, the animation depicts the relationships between points more intuitively. According to [3], learning through animation often outperforms static imagery in helping users understand dynamic relationships. Our animation visually illustrates the correspondence between the written and the target character. Points on the target character move dynamically from their matched locations on the handwritten sample back to their original positions on the target. This motion not only highlights the spatial relationships but also reinforces the understanding of deviations in alignment and structure (See Figure 2).

4 Experiment

On the first day, our preliminary experiment includes preparation, pre-tests, training, and post-tests. On the second day, participants will complete a recognition test, a second post-test, and a survey. Following Connan et al. [7], we schedule the test the day after training, as a week-long interval [6] is too lengthy for a single-session experiment.

We evaluated our VisDev method, labeled as ‘Our’, against the traditional handwriting training method of tracing (‘Tracing’). Prior research [6] shows tracing’s short-term advantages over copying. Therefore, we used tracing as the sole benchmark, reasoning that if VisDev outperforms tracing, it will also exceed copying.

For experiment characters, we selected Japanese Kana and Thai scripts. Kana was chosen due to the study’s location in Japan and its intermediate complexity between Chinese characters and the alphabet, offering a balanced challenge for participants familiar with it. Thai characters were included to assess the method’s impact on learning an unfamiliar script. We used 6 characters each from Hiragana, Katakana (two different forms of Kana), and Thai, displayed in YuGothicLight (Kana) and BoonJot-Regular (Thai) fonts, respectively.

We have 20 right-handed participants (average age: 25.3 years, SD = 5.89, 8 females), all proficient in Japanese Kana but not familiar with Thai. Participants signed informed consent before the experiment and were compensated with a ¥3,000 (approximately \$20) Amazon gift card upon completion. The ethics committee approved the study.

4.1 Experimental Design

Twenty participants were divided into groups A and B, with each learning two sets of characters. Set 1 and Set 2, each comprising three Hiragana, three Katakana, and three Thai characters, were alternately taught using ‘Our’ and ‘Tracing’. Group A learned Set 1 with ‘Our’ and Set 2 with ‘Tracing’, while Group B did the reverse. This crossover design ensured both groups experienced both methods and each character set was practiced using both approaches. Set 1 was always presented first, with methods alternated between groups. No break occurred between the two methods.

4.2 Procedure

The experiment consisted of five steps: **preparation**, **pre-test**, **training**, **post-test**, and **post-test2**. For preparation, participants practiced using six distinct characters: three for ‘Our’ method and three for ‘Tracing’. They are not used in the training or tests. For the pre-test (PRE), they were then shown 18 test characters, each displayed for 6 seconds (1), followed by a 3-second black-screen pause (2), before writing (3) each character three times in a randomized order (see Figure 3). During training with ‘Our’ method, the target character appeared on the right side of the screen along with an alignment grid for stroke positioning. The character disappeared when writing began, promoting memorization. Afterward, participants could assess their accuracy and attempt the character up to five times. For ‘Tracing’, participants traced the target character five times, clearing the written sample between attempts. They are shown in Figure 2. After a 15-minute break, the post-test followed the pre-test procedure, with target characters visible for only 2 seconds to reflect improved recall. The grid was not provided, and the character order remained randomized. The second day began with a recognition test to evaluate retention of the Thai characters, followed by a repeat of the post-test to assess sustained improvements. Participants also completed a survey to share feedback on their experience and provide suggestions: (1) During the test phase (pre and post-test), were you able to write characters as they become the same shape and position as the target characters as much as possible? (Scale, 1 = strongly disagree, 7 = strongly agree); (2) Which of the following practice methods do you think contributes more to your writing ability? (3) What are the reasons for the above choice? (4) Please feel free to write any comments you may have regarding the experiment.

4.3 Evaluation Method

Aesthetic quality is typically assessed through multiple aspects, such as stroke structure, writing speed, and pressure. However, high-level aesthetic assessment can be subjective and requires more advanced evaluation techniques. In our study, we focus on stroke structure to ensure objective and reproducible evaluations of handwriting accuracy. Following previous studies [2, 15, 19], we evaluate handwriting quality and fluency. Writing quality is assessed by structural accuracy and spatial positioning errors. Fluency, reflecting smoothness and consistency, is evaluated through speed, pressure, and the signal-to-noise velocity peaks difference (SNvpd) [7, 8].

Quality: Measured using chamfer distance after aligning the written sample to the target via rigid registration [22] (Structure

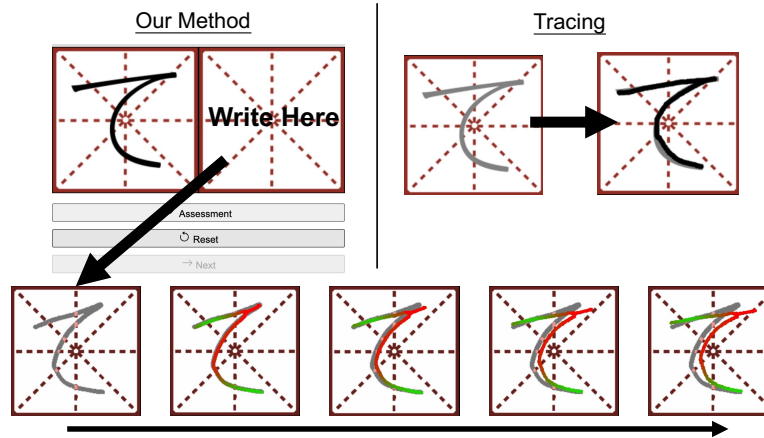


Figure 2: The feedback mode and tracing mode. For the feedback, the matched points will gradually move to the target position.

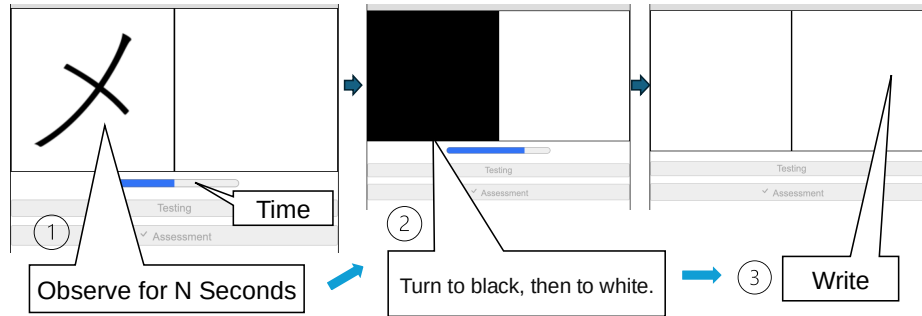


Figure 3: In the testing process, users are given time to view the target character, followed by a brief pause, after which they start writing.

Error); Calculated as the mean distance between points before and after alignment (Relative Position Error).

Fluency: Based on position and timing data (Averaged Velocity); Derived from the difference in velocity peaks after filtering at 5Hz and 10Hz (SNvpd); Average pen pressure recorded for each point (Averaged Pressure).

5 Results

We evaluated the mentioned five metrics, applying a logarithmic transformation to normalize structure and position errors, following [7]. Outliers were removed using the Interquartile Range (IQR) method with cutoffs set to three times the IQR. Regarding the timeline, 'PRE' refers to the measurements before the training, 'POST' denotes the assessment conducted on the first day, and 'POST_2' represents the follow-up test carried out on a subsequent day.

To analyze writing quality improvement relative to 'POST', we conducted paired samples t-tests for structure and relative position errors. ANCOVA was used to compare 'Our' and 'Tracing' methods and the impact of different character sets, with 'PRE' performance as a covariate (baseline performance). Fluency metrics (speed, SNvpd, pressure) were analyzed using the Mann-Whitney U test, as normality was not required. Cliff's Delta measured effect

sizes, and p-values were adjusted using the Benjamini-Hochberg method for reliability.

5.1 Post-test

For the structure error on 'POST', 'Our' method has a significant improvement. 'Tracing' also showed a significant enhancement, though to a lesser degree. The larger Cohen's *d* for 'Our' indicates a more pronounced improvement. The 'Our' method outperformed 'Tracing' significantly, as determined by ANCOVA, which considered initial performance levels. The significance of the differences in improvement between 'Our' and 'Tracing' methods again the 'PRE' scores further confirms the advantage of 'Our' method, with a positive *Cohen's d* underscoring the greater average gain. Similarly, for relative position error on 'POST', 'Our' method showed post-training improvement significantly. 'Tracing' method also resulted in significant improvement, though less so. ANCOVA results again favored the 'Our' method. However, the difference in the improvement gaps is not very significant, even though Cohen's *d* is not much smaller than when assessing structure errors. Both methods enhanced writing quality, with the 'Our' method showing greater improvement over 'Tracing'. According to Cohen's *d*, the enhancement in structure error surpassed that in relative position

error, indicating a more significant improvement in the fundamental aspects of character formation. The detailed data is presented in Table 1. For speed, both ‘Our’ and ‘Tracing’ methods showed a decrease in mean values post-training, yet the Mann–Whitney U test revealed no significant differences, with all p-values exceeding 0.05 and small effect sizes indicated by Cliff’s Delta. Similarly, SNvpd scores showed no significant change post-training, with p-values remaining above 0.05 and Cliff’s Delta values indicating negligible effect sizes. Pressure metrics also indicated no significant differences after training, with p-values above 0.05 and Cliff’s Delta reflecting larger effect sizes compared with SNvpd and speed. The detailed data can be found in Table 2.

ANCOVA found no significant differences across character sets, with ‘POST’ performance greatly impacting results. The ANCOVA analysis comparing results before and after training revealed no significant differences in performance across character sets for both structure error and relative position error (Structure error: $F(1, 355) = 2.44, p > .1$; Relative Position Error: $F(1, 355) = 0.00, p > .5$). Additionally, there was no significant interaction between the method and character sets (Structure error: $F(1, 355) = 1.88, p > .1$; Relative Position Error: $F(1, 355) = 1.14, p > .05$), indicating that the effectiveness of both methods was consistent across different character sets. The pre-test performance, serving as a covariate, had a significant impact on the outcomes for both metrics (Structure error: $F(1, 355) = 300.0, p < .001$; Relative Position Error: $F(1, 355) = 139.6, p < .001$). This underscores the importance of initial performance levels in explaining variations in learning outcomes, highlighting that the participants’ starting proficiency plays a crucial role in the effectiveness of handwriting training.

5.2 Post-test2

For the structure error on ‘POST_2’, both methods continued to show significant improvements in writing quality. The ‘Our’ method showed a change from ‘PRE’ to ‘POST_2’, with a significant difference. Similarly, ‘Tracing’ improved from ‘PRE’ to ‘POST_2’ with a significant difference. However, the comparison between the two methods showed no significant difference. For relative position error, neither method demonstrated significant improvement from ‘PRE’ to ‘POST_2’. The difference in improvement between the methods was also not significant. This outcome aligns with prior findings [6], suggesting that for ‘POST_2’, there is no significant difference in performance between the methods. Refer to Table 3 for the detailed data. For speed, both ‘Our’ and ‘Tracing’ methods showed no significant change in performance from before training to ‘POST_2’ on the following day, with Mann–Whitney U test results indicating all p-values were greater than 0.05. Similarly, for SNvpd, no significant differences were observed between the ‘PRE’ and ‘POST_2’ measurements for either method. Pressure measurements also remained statistically unchanged from ‘PRE’ to ‘POST_2’ in both groups, further underscoring the lack of significant difference. Refer to Table 2 for the detailed data.

5.3 User Feedback

‘Our’ method was praised for its valuable feedback and memorization focus, with 8 participants noting its benefits. Tracing was appreciated for its simplicity and familiarity, with 8 participants favoring

it for ease of use. Preferences were nearly evenly split, highlighting varying learning preferences. Most participants (13/20) correctly identified Thai characters in the recognition task, showing no significant differences between methods. Confidence in writing quality improved for most participants, as reflected in self-evaluations.

6 Discussion

‘Our’ method outperformed ‘Tracing’ in all metrics, however more in structural error than relative position error. This might be due to differences between training and testing environments: a grid (Figure 2) was used during training but not during testing (Figure 3), likely impacting relative position accuracy. While grids are commonly used in traditional handwriting instruction, our results suggest that such aids may be counterproductive in broader writing scenarios.

ANCOVA revealed that pre-test scores significantly influenced final scores, suggesting that initial writing ability outweighs the effect of the training method. This underscores the importance of accounting for baseline skill levels when evaluating handwriting training. Accordingly, training programs should be tailored: advanced users might benefit from fewer sessions or stricter criteria. On day two, both methods improved writing quality, but the gains were smaller than on day one. Partial skill loss was observed, rather than total forgetting, and the differences between methods diminished—likely due to the overriding effect of baseline skills. Habitual tendencies led participants to revert to familiar patterns, causing outcomes to converge. In longer-term studies, methods may yield similar results, suggesting the focus should shift to achieving equivalent outcomes more efficiently. Higher initial scores indicate stronger natural copying skills, which often lead to fewer iterations and better final performance. This relationship can help predict outcomes and guide training adjustments, ensuring more efficient, individualized learning.

In terms of fluency, evaluated through SNvpd, speed, and pressure, no significant differences were found, indicating that fluency remained stable despite training. Adult handwriting fluency may benefit from well-developed motor skills, allowing individuals to integrate feedback without losing fluidity.

Our findings also showed no significant performance differences across characters in the assessments. This may be due to allowing participants to preview the test characters, reducing the impact of unfamiliarity. As a result, users achieved consistent spatial accuracy even with characters in an unfamiliar font.

7 Future Work and Practical Implementation

Our study focused on a limited set of Thai and Japanese characters due to constraints in our registration method. Future work will refine this approach to accommodate a wider range of characters, such as Chinese. Currently, all participants were right-handed. We aim to include a more diverse group in future studies, including left-handed individuals and those with varied linguistic and educational backgrounds. The experiment spanned two days, but future studies will extend over several months to examine the long-term effects of handwriting training. Regarding comparison methods, we did not include copying as a baseline, as our primary goal was to validate our approach in a simplified setting. Future studies will

Table 1: Summary of statistical results for quality improvements, including metrics (e.g., Structure Error, Position Error), conditions (PRE and POST), mean values (M), standard deviations (SD), T -values (paired comparisons), F -values (ANCOVA), p -values (statistical significance), and Cohen’s d (effect size).

Metric	Condition	M	SD	Test Statistics
Structure Error (Our)	PRE	-3.07	0.39	$T(179) = 5.87, p < .001, d = 0.36$
	POST	-3.21	0.37	
Structure Error (Tracing)	PRE	-3.10	0.38	$T(179) = 2.59, p < .05, d = 0.15$
	POST	-3.15	0.36	
ANCOVA	-	-	-	$F(1, 355) = 6.67, p < .05$
Improvement Difference Comparison	-	-	-	$T(358) = 2.58, p < .05, d = 0.27$
Position Error (Our)	PRE	-2.70	0.45	$T(179) = 5.69, p < .001, d = 0.39$
	POST	-2.86	0.44	
Position Error (Tracing)	PRE	-2.71	0.39	$T(179) = 2.73, p < .005, d = 0.21$
	POST	-2.79	0.39	
ANCOVA	-	-	-	$F(1, 355) = 5.18, p < .05$
Improvement Difference Comparison	-	-	-	$T(358) = 2.18, p = .06, d = 0.23$

Table 2: Summary of results for Speed, SNvpd, and Pressure before training (PRE), POST, and POST_2, including mean values (M), standard deviations (SD), and Cliff’s Delta (effect size).

PRE, POST						PRE, POST_2					
Metric	Method	Condition	M	SD	Cliff's Δ	Metric	Method	Condition	M	SD	Cliff's Δ
Speed	Our	PRE	215.79	86.90	0.023	Speed	Our	PRE	215.79	86.90	0.011
		POST	207.57	69.64				POST_2	208.93	66.11	
	Tracing	PRE	222.92	96.85	0.016		Tracing	PRE	222.92	96.85	0.019
		POST	214.58	77.90				POST_2	213.52	72.45	
SNvpd	Our	PRE	8.94	5.08	0.010	SNvpd	Our	PRE	8.94	5.08	0.010
		POST	8.96	5.32				POST_2	8.66	4.37	
	Tracing	PRE	8.80	5.83	0.004		Tracing	PRE	8.80	5.83	-0.053
		POST	8.95	6.39				POST_2	8.92	5.03	
Pressure	Our	PRE	0.19	0.13	-0.066	Pressure	Our	PRE	0.19	0.13	-0.036
		POST	0.20	0.15				POST_2	0.18	0.10	
	Tracing	PRE	0.18	0.13	-0.120		Tracing	PRE	0.18	0.13	-0.070
		POST	0.21	0.16				POST_2	0.18	0.11	

Table 3: Summary of results for Structure Error and Relative Position Error before training (PRE) and POST_2, including mean values (M), standard deviations (SD), T -values (paired comparisons), p -values (statistical significance), and Cohen’s d (effect size).

Metric	Method	Condition	M	SD	Test Statistics
Structure Error	Our	PRE	-3.07	0.39	$T(179) = 3.85, p < .001, d = 0.24$
		POST_2	-3.17	0.35	
	Tracing	PRE	-3.10	0.38	$T(179) = 3.01, p < .001, d = 0.19$
		POST_2	-3.17	0.36	
Improvement Diff. Comparison	-	-	-	-	$T(358) = 0.66, p > .1, d = 0.07$
Relative Position Error	Our	PRE	-2.70	0.45	$T(179) = 2.05, p > .05, d = 0.14$
		POST_2	-2.76	0.45	
	Tracing	PRE	-2.71	0.39	$T(179) = 1.77, p > .05, d = 0.14$
		POST_2	-2.77	0.45	
Improvement Diff. Comparison	-	-	-	-	$T(358) = 0.12, p > .1, d = 0.01$

incorporate copying as a control to explore its role alongside tracing in handwriting learning. Additionally, we plan to develop broader aesthetic evaluation criteria, integrating stroke order and writing pressure.

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